

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286090

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

KRISHNADASS; Jayakumar et al.

ELECTROCHEMICAL CELL SYSTEMS INCLUDING FLUID EJECTOR

Abstract

An electrochemical cell system includes a stack of electrochemical cells, and a fluid ejector configured to mix a first fluid stream and a second fluid stream to form a mixed fluid stream that is provided to the stack. The fluid ejector includes a suction chamber having a secondary inlet configured to receive the second fluid stream, an inlet nozzle configured to inject the first fluid stream into the suction chamber, a mixing chamber fluidly connected to the suction chamber and configured to mix the first fluid stream and the second fluid stream to form the mixed fluid stream, and a diffusion chamber fluidly connected to the mixing chamber and having an outlet that is fluidly connected to the stack.

Inventors: KRISHNADASS; Jayakumar (Santa Clara, CA), WEINGAERTNER; David (Sunnyvale, CA), KANDASWAMY; Easwaramoorthy Vellakinar (Fremont, CA)

Applicant: BLOOM ENERGY CORPORATION (San Jose, CA)

Family ID: 96949875

Appl. No.: 19/051752

Filed: February 12, 2025

Related U.S. Application Data

us-provisional-application US 63563708 20240311

Publication Classification

Int. Cl.: H01M8/04082 (20160101); C25B9/77 (20210101); C25B15/08 (20060101); H01M8/04089 (20160101); H01M8/04111 (20160101); H01M8/04119 (20160101); H01M8/12 (20160101); H01M8/2457 (20160101); H01M8/2484 (20160101); H01M8/249 (20160101)

U.S. Cl.:

CPC **H01M8/04201** (20130101); **C25B9/77** (20210101); **C25B15/08** (20130101);
H01M8/04097 (20130101); **H01M8/04111** (20130101); **H01M8/04164** (20130101);
H01M8/2457 (20160201); **H01M8/2484** (20160201); **H01M8/249** (20130101);
H01M2008/1293 (20130101)

Background/Summary

FIELD

[0001] Aspects of the present invention relate to electrochemical cell systems, and more particularly, to solid oxide fuel cell and electrolyzer cell systems that include fluid ejectors.

BACKGROUND

[0002] Fuel cells, such as solid oxide fuel cells, are electrochemical devices which can convert energy stored in fuels to electrical energy with high efficiency. High temperature fuel cells include solid oxide and molten carbonate fuel cells. These fuel cells may operate using hydrogen and/or hydrogen-containing fuels. There are classes of fuel cells, such as the solid oxide regenerative fuel cells, that also allow reversed operation, such that oxidized fuel can be reduced back to unoxidized fuel using electrical energy as an input.

SUMMARY

[0003] According to various embodiments, an electrochemical cell system includes a stack of electrochemical cells; a first fluid ejector configured to mix a first fluid stream and a second fluid stream to form a mixed fluid stream that is provided to the stack, the first fluid ejector comprising: a suction chamber comprising a secondary inlet configured to receive the second fluid stream; an inlet nozzle configured to inject the first fluid stream into the suction chamber; a mixing chamber fluidly connected to the suction chamber and configured to mix the first fluid stream and the second fluid stream to form the mixed fluid stream; and a diffusion chamber fluidly connected to the mixing chamber and comprising an outlet that is fluidly connected to the stack.

[0004] According to various embodiments, a method of operating an electrochemical cell system comprises providing a first fluid stream to an inlet nozzle of a fluid ejector; providing a second fluid stream from a stack of electrochemical cells to a second inlet located in a suction chamber of the fluid ejector; mixing the first fluid stream and the second fluid stream in a mixing chamber of the fluid ejector to form a mixed fluid stream; and providing the mixed fluid stream from a diffusion chamber of the fluid ejector to the stack.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate example embodiments of the invention, and together with the general description given above and the detailed description given below, serve to explain the features of the invention.

[0006] FIG. 1A is a perspective view of a stack of electrochemical cells, according to various embodiments of the present disclosure, and FIG. 1B is a sectional view of a portion of the stack of FIG. 1A.

[0007] FIG. 2 is a schematic view of a fuel cell fuel cell system, according to various embodiments of the present disclosure.

[0008] FIG. 3 is a schematic diagram showing a process flow in an electrolyzer system 300,

according to various embodiments of the present disclosure.

[0009] FIG. 4 is a schematic diagram of a fluid ejector that may be utilized in the systems of FIGS. 2 and 3, according to various embodiments of the present disclosure.

[0010] FIG. 5A is a schematic diagram showing a process flow in a SOFC system, according to various embodiments of the present disclosure.

[0011] FIGS. 5B-5D are schematic diagrams showing process flows of alternative SOFC systems, according to various embodiments of the present disclosure.

[0012] FIG. 6A is a schematic diagram showing a process flow in a SOEC system, according to various embodiments of the present disclosure.

[0013] FIGS. 6B and 6C are schematic diagrams showing process flows of alternative SOEC systems, according to various embodiments of the present disclosure.

[0014] FIG. 7A is a schematic diagram showing a process flow in a SOEC system, according to various embodiments of the present disclosure.

[0015] FIGS. 7B-7D are schematic diagrams showing process flows of alternative SOEC systems, according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0016] As set forth herein, various aspects of the disclosure are described with reference to the exemplary embodiments and/or the accompanying drawings in which exemplary embodiments of the invention are illustrated. This invention may, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiments shown in the drawings or described herein. It will be appreciated that the various disclosed embodiments may involve particular features, elements or steps that are described in connection with that particular embodiment. It will also be appreciated that a particular feature, element or step, although described in relation to one particular embodiment, may be interchanged or combined with alternate embodiments in various non-illustrated combinations or permutations.

[0017] The various embodiments will be described in detail with reference to the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. References made to particular examples and implementations are for illustrative purposes and are not intended to limit the scope of the invention or the claims.

[0018] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, examples include from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about” or “substantially” it will be understood that the particular value forms another aspect. In some embodiments, a value of “about X” may include values of $\pm 1\%$ X. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0019] In a high temperature fuel cell system, such as a solid oxide fuel cell (SOFC) system, an oxidizing flow is directed to the cathode (i.e., air) side of the fuel cell while a fuel flow is directed to the anode (i.e., fuel) side of the fuel cell. The oxidizing flow is typically air, while the fuel flow can be a hydrogen (H_{2}) or a hydrocarbon fuel, such as methane, natural gas, propane (LPG), ethanol, or methanol, or another suitable fuel, such as ammonia. The fuel cell, operating at a typical temperature between $700^{\circ}C.$ and $950^{\circ}C.$, enables the transport of negatively charged oxygen ions from the cathode flow stream to the anode flow stream, where the oxygen ions combine with either free hydrogen or hydrogen in a hydrocarbon or ammonia molecule to form water vapor and optionally combine with carbon monoxide to form carbon dioxide. The excess electrons from the negatively charged ions are routed back to the cathode side of the fuel cell through an electrical circuit completed between anode and cathode, resulting in an electrical current flow through the circuit.

[0020] Solid oxide electrolyzer cells (SOEC) produce hydrogen and oxygen from water. In a SOEC, a positive potential is applied to the air side of the cell and the oxygen ions are transported

from the fuel side to the air side. Since the cathode and anode are reversed between SOFC and SOEC (i.e., SOFC cathode is SOEC anode, and SOFC anode is SOEC cathode), going forward, the SOFC cathode (SOEC anode) will be referred to as the air electrode, and the SOFC anode (SOEC cathode) will be referred to as the fuel electrode. During SOEC operation, water (e.g., steam) in the fuel stream is reduced ($\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + \text{O}^{2-}$) to form H_2 gas and O^{2-} ions, O^{2-} ions are transported through the solid electrolyte, and then oxidized on the air side (O^{2-} to O_2) to produce molecular oxygen.

[0021] FIG. 1A is a perspective view of an electrochemical cell stack **100**, and FIG. 1B is a side cross-sectional view of a portion of the stack **100** of FIG. 1A. Referring to FIGS. 1A and 1B, the stack **100** includes multiple electrochemical cells **1** that are separated by interconnects **10**, which may also be referred to as gas flow separator plates or bipolar plates. The electrochemical cells **1** may be fuel cells or electrolyzer cells. Accordingly, the stack **100** may be referred to as a solid oxide fuel cell (SOFC) stack **100** or a solid oxide electrolyzer cell (SOEC) stack **100**. Each electrochemical cell **1** includes an air electrode **3**, a solid oxide electrolyte **5**, and a fuel electrode **7**. The stack **100** also includes optional internal fuel riser channels **22**.

[0022] Each interconnect **10** electrically connects adjacent electrochemical cells **1** in the stack **100**. In particular, an interconnect **10** may electrically connect the fuel electrode **7** of one electrochemical cell **1** to the air electrode **3** of an adjacent electrochemical cell **1**. FIG. 1B shows that the lower electrochemical cell **1** is located between two interconnects **10**.

[0023] Each interconnect **10** includes fuel ribs **12A** that at least partially define fuel channels **8A** and air ribs **12B** that at least partially define air channels **8B**. The interconnect **10** may operate as a gas-fuel separator that separates a fuel (e.g., hydrogen for a SOFC or steam for SOEC) flowing to the fuel electrode **7** of one electrochemical cell **1** in stack **100** from oxidant, such as air, flowing to the air electrode **3** of an adjacent electrochemical cell **1** in stack **100**. At either end of the stack **100**, there may be an air end plate or fuel end plate (not shown) for providing air or fuel, respectively, to the end electrode.

[0024] An electrochemical cell column may include one stack **100** or multiple stacks **100** arranged on one another. The column may be internally or externally manifolded for fuel and/or air. Optional anode splitter plates may be disposed between adjacent stacks **100** to provide fuel to the cells of each stack **100** as described in U.S. Pat. No. 10,511,047 B2, which is incorporated herein by reference in its entirety.

[0025] While a co-flow or counter-flow interconnect **10** is illustrated in FIG. 1B, in alternative embodiments, the interconnect **10** may comprise a crossflow interconnect in which the air and fuel channels extend perpendicular to each other, as described in U.S. Pat. No. 11,355,762 B2, which is incorporated herein by reference in its entirety. For example, such interconnects **10** may include two or more fuel holes per side of the interconnect.

[0026] FIG. 2 is a schematic diagram of a solid oxide fuel cell system **200** that includes a power module **202** that uses hydrogen fuel, according to various embodiments of the present disclosure.

[0027] The power module **202** contains a hotbox **101**. One or more electrochemical cell stacks **100** (or columns of stacks) are located in the hotbox **101**. In one embodiment, the stacks **100** comprise solid oxide fuel cell stacks.

[0028] The power module **202** may also contain an anode recuperator **120** heat exchanger, a cathode recuperator **130** heat exchanger, and a startup heater **150**. In some embodiments, the power module **202** may optionally include an anode exhaust cooler **140**. The power module **202** may also include an air blower **160** (e.g., system blower), which may be disposed outside of the hotbox **101**. However, the present disclosure is not limited to any particular location for each of the module components with respect to the hotbox **101**.

[0029] The anode recuperator **120** receives fuel (e.g., H_2) from a fuel inlet (e.g., a hydrogen vessel and/or hydrogen line) **102** through a fuel inlet conduit **112**. The fuel is heated in the anode recuperator **120** by fuel exhaust (e.g., anode exhaust) output from the stack **100**, before being

provided to the stack **100** by a stack fuel conduit **113**. A first heater conduit **152A** may fluidly connect the fuel inlet **102** to the startup heater **150**. An optional second heater conduit **152B** may also fluidly connect the fuel inlet **102** to the startup heater **150**. Accordingly, the startup heater **150** may receive fuel provided by either or both of the first and second heater conduits **152A**, **152B**. The two inlets provide circumferential uniformity to the heaters **150**. Alternatively, there may be only one heater conduit **152A** or more than two heater conduits (e.g., three or four heater conduits). The conduits **112**, **152A** and **152B** may be fluidly connected to the fuel inlet **102** using any suitable fluid connectors. For example, the fuel inlet conduit **112** may be connected to the fuel inlet **102**, the first heater conduit **152A** may be connected to the fuel conduit **112** at a first two way splitter downstream of the fuel inlet **102**, and the second heater conduit **152B** may be connected to the first heater conduit **152A** at a second two way splitter downstream of the first two way splitter. Alternatively, a single three way splitter may split fuel from the fuel inlet **102** into all three conduits **112**, **152A** and **152B**. Other fluid connections may also be used to connect the fuel inlet **102** to the three conduits **112**, **152A** and **152B**. The first and second heater conduits **152A**, **152B** may be connected to the same or different fuel inlets of the startup heater **150**. For example, the startup heater **150** may include a heating fuel inlet **154A** and/or an ignition fuel inlet **154B** connected to respective heater conduits **152A** and **152B**.

[0030] The startup heater **150** may also receive air exhaust (i.e., cathode exhaust) output from the stack **100** through an exhaust conduit **164A**. The startup heater **150** may include a fuel oxidation catalyst (e.g., a noble metal catalyst) and/or heating element (e.g., resistive and/or radiative heating element). The heater **150** may generate heat by catalytically and/or thermally oxidizing received fuel using the air exhaust. Exhaust output from the startup heater **150** may be provided to the cathode recuperator **130** through exhaust conduit **164B**. Exhaust output from the cathode recuperator **130** may be exhausted from the hotbox **101** through exhaust conduit **164C** and exhaust outlet **132** in the hotbox **101**. An exhaust conduit **164D** may be configured to receive exhaust output from the exhaust outlet **132**. In some embodiments, the exhaust conduit **164D** may be part of or connected to a recycling manifold **110** configured to receive exhaust output from multiple hotboxes **100**, as discussed in more detail below.

[0031] The air blower **160** may be configured to provide air (e.g., an air inlet stream) to the anode exhaust cooler **140** through air conduit **162A**. Air flows from the anode exhaust cooler **140** to the cathode recuperator **130** through air conduit **162B**. The air is heated in the cathode recuperator **130** by the air exhaust output from the stack **100** (or by startup heater **150** exhaust output if the fuel is also provided to the startup heater **150**, where the fuel is oxidized by the air exhaust to form the oxidized fuel heater exhaust output). The heated air flows from the cathode recuperator **130** to the stack **100** through air conduit **162C**.

[0032] Fuel exhaust (e.g., an anode exhaust stream generated in the stack **100**) is provided to the anode recuperator **120** through fuel exhaust conduit **114A**. The fuel exhaust may contain unreacted hydrogen fuel and water. Fuel exhaust output from the anode recuperator **120** may be provided to a fuel exhaust outlet **104** of the hotbox **101**, by fuel exhaust conduit **114B**. In some embodiments, the optional anode exhaust cooler **140** may be configured to cool the fuel exhaust flowing through the fuel exhaust conduit **114B** by the inlet air stream from the air conduit **162A**, prior to the fuel exhaust reaching the fuel exhaust outlet **104**. The power module **202** may also optionally include a purge conduit **244** that fluidly connects the first heater conduit **152A** to the fuel exhaust conduit **114B**.

[0033] The power module **202** may further comprise a system controller **125** configured to control various elements of the module **202** and communicate with other components connected to the power module **202**. The controller **125** may include a central processing unit configured to execute stored instructions. For example, the controller **125** may be configured to control the air flow through the power module **202** and to open and close the fuel flow to the power module **202**.

[0034] In some embodiments, the fuel cell stacks **100** may be arranged in the hotbox **101** around a

central column including the anode recuperator **120**, the startup heater **150**, and the optional anode exhaust cooler **140**. In particular, the anode recuperator **120** may be disposed radially inward of the startup heater **150**, and the anode exhaust cooler **140** may be mounted over the anode recuperator **120** and the startup heater **150**.

[0035] The system **200** may include a condenser **180** and a fluid ejector **400**. In particular, the condenser **180** may be used to condense water vapor from anode exhaust output from the anode exhaust cooler **140** via the recycling manifold **110**. The recycling manifold **110** may receive anode exhaust output from the hotbox **202** and provide the anode exhaust to the condenser **180** and then to a secondary inlet of the fluid ejector **400**. As discussed in more detail below with regard to FIG. **4**, the fluid ejector **400** is configured to mix the fresh fuel received at its primary inlet from the fuel source **102** with anode exhaust received at its secondary inlet from the recycling manifold **110** to form the fuel inlet stream (e.g., a mixed stream), which is provided to the anode recuperator **120** via the fuel inlet conduit **112** and then to the stack **100** via the stack fuel conduit **113**. In one embodiment, the condenser **180** and the ejector **400** may be located outside the housing of the power module **202**. An optional downstream pressure regulator **401** may be located downstream of the ejector **400** on the fuel inlet conduit **112**. When the stack **100** current drops, the fuel flow through the ejector **400** will also drop, and the outlet pressure may become too high. The downstream pressure regulator **401** controls the outlet pressure from the ejector **400** to remain in a desired range. An optional upstream pressure regulator **62** may be located upstream of the ejector **400** on the fuel inlet conduit **112**. The upstream pressure regulator **62** may be omitted if the system site fuel pressure is controlled to be in a desired range.

[0036] FIG. **3** is a schematic diagram of a solid oxide electrolyzer cell system **300** that includes a hydrogen generation module **302** that generates hydrogen from water, according to various embodiments of the present disclosure. The hydrogen generation module **302** may include a hotbox **301** containing one or more SOEC stacks **100** or columns, including multiple SOECs, as described above with respect to FIGS. **1A** and **1B**.

[0037] The system **300** may also include a steam generator **304**, a steam recuperator heat exchanger **308**, an air recuperator heat exchanger **312**, an air blower **318**, an air heater **355**, a stack heater **360**, and a steam heater **370**. The system **300** may also optionally include at least one of an air pre-heater heat exchanger **354**, a water preheater heat exchanger **302**, a mixer **306**, a hydrogen processor **320** and/or a hydrogen flow divider (e.g., splitter or valve) **322**.

[0038] The system **300** may include a hotbox **301** that houses various components, such as the stack **100**, the steam recuperator **308**, the air recuperator **312**, an air heater **355**, a stack heater **360** and a steam heater **370**. In some embodiments, the hotbox **301** may include multiple stacks **100** or multiple columns of stacks **100**. The water preheater **302** and the steam generator **304** may be located external to the hotbox **301**, as shown in FIG. **3**. Alternatively, the water preheater **302** and/or the steam generator **304** may be located inside the hotbox **301**. In another alternative embodiment, the system **300** may utilize an external steam source, in which case the water preheater **302** and the steam generator **304** may be omitted.

[0039] During operation, the stack **100** may be provided with steam (e.g., steam inlet stream) and electric power (e.g., current or voltage) from an external power source. In particular, the steam may be provided to the fuel electrodes **7** of the electrolyzer cells **1** of the stack **100**, and the power source may apply a voltage between the fuel electrodes **7** and the air electrodes **3**, in order to electrochemically split water (e.g., steam) molecules and generate hydrogen and oxygen. Air may also be provided to the air electrodes **3**, in order to sweep the oxygen from the air electrodes **3**. As such, the stack **100** may output a hydrogen stream and an oxygen-rich exhaust stream, such as an oxygen-rich air stream ("oxygen exhaust stream").

[0040] In order to generate the steam, water may be provided to the system **300** from a water source **350**. The water source **350** may include a municipal water supply (e.g., water pipe) and/or a water storage tank. The water may be deionized (DI) water that is deionized as much as is practical

(e.g., $<0.1 \mu\text{S}/\text{cm}$), in order to prevent and/or minimize scaling during vaporization. In some embodiments, the water source **350** may include one or more deionization beds (e.g., downstream of the water pipe or tank). The water source **350** may provide the water to the system **300** via a water inlet conduit **250**. In various embodiments, the water inlet conduit **250** may include a water flow control device **251** such as a valve, a mass flow controller, a positive displacement pump, a water flow meter, or the like, in order to provide a desired water flow rate to the system **300**.

[0041] If the system **300** includes the water preheater **302**, the water may be provided from the water source **350** to the water preheater **302** through the water inlet conduit **250**. The water preheater **302** may be a heat exchanger configured to heat the water using heat recovered from the oxygen exhaust stream from the stack **100**. Preheating the water may reduce the total power consumption of the system **300** per unit of hydrogen generated. In particular, the water preheater **302** may recover heat from the oxygen exhaust stream that may not be recoverable by the air recuperator **312**, as discussed below. The water preheater may heat the water to a temperature above 50°C ., such as a temperature of about 70°C . to 80°C . The oxygen exhaust stream may be output from the water preheater **302** at a temperature above 80°C ., such as above 100°C ., such as a temperature of about 120°C . to 140°C .

[0042] The water output from the water preheater **302** (or from the water source **350** if the water preheater **302** is omitted) may be provided to the steam generator **304** through a water conduit **202**. The steam generator **304** may be configured to heat the water to convert the water into steam. The steam generator **304** may include a heating element to vaporize the water and generate steam. For example, the steam generator **304** may include an AC or DC resistance heating element or an induction heating element. Alternatively, the steam generator **304** may comprise a heat exchanger which is located inside the hotbox **301** and which is heated by one or more hot exhaust streams flowing through the hotbox **301**. The steam generator **304** heats the water above 100°C . to generate steam, such as a temperature of about 120°C . to 145°C .

[0043] The steam generator **304** may include multiple zones/elements that may or may not be mechanically separate. For example, the steam generator **304** may include a pre-boiler to heat the water up to or near to the boiling point. The steam generator **304** may also include a vaporizer configured to convert the pre-boiled water into steam. The steam generator **304** may also include a deaerator to provide a relatively small purge of steam to remove dissolved air from the water prior to bulk vaporization. The steam generator **304** may also include an optional superheater configured to further increase the temperature of the steam generated in the vaporizer. The steam generator **304** may include a demister pad located downstream of the heating element and/or upstream from the super heater. The demister pad may be configured to minimize entrainment of liquid water in the steam output from the steam generator **304** and/or provided to the superheater.

[0044] If the steam product is superheated, it will be less likely to condense downstream from the steam generator **304** due to heat loss to ambient conditions. Avoidance of condensation is preferable, as condensed water is more likely to form slugs of water that may cause significant variation of the delivered mass flow rate with respect to time. It may also be beneficial to avoid excess superheating, in order to limit the total power consumption of the system **300**. For example, the steam may be superheated by an amount ranging from about 10°C . to about 100°C .

[0045] In some embodiments, a small amount of liquid water (e.g., from about 0.5% to about 2% of incoming water) may be periodically or continuously discharged from the steam generator **304** via a liquid discharge conduit **224**. In particular, the discharged liquid water may include scale and/or other mineral impurities that may accumulate in the steam generator **304** while vaporizing water to generate steam. Therefore, this discharged liquid water is not desirable for being recycled into the water inlet stream from the water source **350**. This liquid discharge may be mixed with the hot oxygen exhaust stream output from the water preheater **302** into an exhaust conduit **205**. If the hot oxygen exhaust stream has a temperature above 100°C ., such as 120 to 140°C ., the liquid water discharge may be evaporated by the hot oxygen exhaust stream, such that no liquid water is

required to be discharged from the system **300**. The system **300** may optionally include a water pump **324** configured to pump and regulate the liquid water discharge in the liquid discharge conduit **224** output from the steam generator **304** into the exhaust conduit **205** from the water preheater **302**. Optionally, a flow regulator, such as proportional solenoid valve, may be added to the liquid discharge conduit **224** in addition to the pump **324** to additionally regulate the flow of the liquid water discharge.

[0046] Blowdown from the steam generator **304** may be beneficial for long term operation, as the water will likely contain some amount of mineralization after deionization. Typical liquid blowdown may be on the order of 1%. The blowdown may be continuous, or may be intermittent, e.g., ten times the steady state flow for 6 seconds out of every minute, five times the steady state flow for 1 minute out of every 5 minutes, etc. The need for a water discharge stream can be eliminated by pumping the blowdown into the hot oxygen exhaust. In this case, the pump **324** and liquid discharge conduit **224** may be omitted.

[0047] The steam output from the steam generator **304** may be provided to the steam recuperator **308** via a steam conduit **204**. However, if the system **300** includes the optional mixer **306**, the steam may be provided to the mixer **306** prior to being provided to the steam recuperator **308**. In particular, the steam may include small amounts of dissolved air and/or oxygen. The mixer **306** may be configured to mix the steam with hydrogen gas, in order to maintain a reducing environment in the stack **100**, and in particular, at the fuel electrodes **7**.

[0048] The mixer **306** may be configured to mix the steam with hydrogen received from a hydrogen storage device (e.g., hydrogen storage vessel) **352** and/or with a portion of the hydrogen and stream recycle stream output from the stack **100**. The hydrogen addition rate may be set to provide an amount of hydrogen that exceeds an amount of hydrogen needed to react with an amount of oxygen dissolved in the steam. The hydrogen addition rate may either be fixed or set to a constant water to hydrogen ratio. However, if the steam is formed using water that is fully deaerated, the mixer **306** and/or hydrogen addition into the steam may optionally be omitted.

[0049] In some embodiments, the hydrogen may be provided to the mixer **306** during system startup and shutdown modes, and optionally during steady-state operation modes. For example, during the startup and shutdown modes (or other modes where the system **300** is not generating hydrogen, such as a fault mode), the hydrogen may be provided to the mixer **306** from the hydrogen storage device **352** via a stored hydrogen conduit **252**.

[0050] During the steady-state operating mode, the hydrogen flow from the hydrogen storage device **352** may be stopped (e.g., by shutting off the outlet valve from the hydrogen storage device). Instead, a first portion of a hydrogen exhaust stream (e.g., the hydrogen and steam product steam) generated by the stack **100** is diverted to the ejector **400** through the hydrogen recycle manifold **210A**. In particular, the system **300** may include a hydrogen flow divider **322**, such as a splitter and/or valve, configured to selectively divert a portion of the hydrogen exhaust stream flowing through the hydrogen product manifold **210** to the secondary entrance of the ejector **400** via the hydrogen recycle manifold **210A** during the steady-state mode operation.

[0051] The mixed steam and hydrogen inlet stream is provided from the mixer **306** through the primary entrance of the ejector **400** into a steam recuperator heat exchanger **308** via a steam and hydrogen conduit **206**. The mixed steam and hydrogen inlet stream in conduit **206** may have a temperature above 100° C., such as 120° C. to 140° C. The mixed steam and hydrogen inlet stream is heated in the steam recuperator **308** by the hydrogen exhaust (i.e., the hydrogen and steam product stream) provided from the stack **100**. The hydrogen exhaust may be provided from the stack **100** to the steam recuperator **308** via a hydrogen outlet conduit **209**. The heated mixed steam and hydrogen inlet stream is provided from the steam recuperator heat exchanger **308** into the fuel side inlet of the stack **100** via the fuel inlet conduit **208**. The mixed steam and hydrogen inlet stream in the fuel inlet conduit **208** may have a temperature above 500° C., such as 550° C. to 600° C. In an alternative embodiment, the hydrogen flow from the hydrogen storage device **352** may be

provided downstream of the ejector **400**. In this alternative embodiment, the mixer **306** may be located downstream of the ejector **400** and the stored hydrogen conduit **352** is connected to the mixer **306** downstream of the ejector **400**.

[0052] The hydrogen exhaust is output from the hotbox **301** (e.g., from the steam recuperator **308** and/or the optional air preheater **354**) into the hydrogen product manifold **210** at a temperature of 150° C. to 250° C. A second portion of the hydrogen exhaust that is not diverted by the hydrogen flow divider **322** into the ejector **400** via the hydrogen recycling manifold **210A** continues through the downstream portion **210B** of the hydrogen product manifold **220** into the hydrogen processor **320**. The hydrogen exhaust may be compressed and/or purified in the hydrogen processor **320**. The hydrogen processor **320** may include a high temperature hydrogen pump that operates at a temperature of from about 320° C. to about 200° C., in order to capture from about 70% to about 90% of the hydrogen from the hydrogen exhaust. The removed hydrogen is stored and/or provided for one or more end uses. In one embodiment, the hydrogen processor **320** includes an electrochemical hydrogen pump, a liquid ring compressor, a diaphragm compressor or combination thereof. For example, the hydrogen processor may include a series of electrochemical hydrogen pumps, which may be disposed in series and/or in parallel with respect to a flow direction of the hydrogen exhaust, in order to compress the hydrogen exhaust. The final product from compression may still contain traces of water. As such, the hydrogen processor **320** may optionally include a dewatering device, such as a condenser, a temperature swing adsorption reactor or a pressure swing adsorption reactor, to remove this residual water, if necessary.

[0053] The air recuperator heat exchanger **312** may be provided with ambient air by an air blower **318** via an air inlet conduit **218** and an optional preheated air conduit **254**. The oxygen exhaust output from the stack **100** may be provided to the air recuperator **312** via an oxygen outlet conduit **222**. The air recuperator **312** may be configured to heat the air using heat extracted from the stack oxygen exhaust (i.e., the oxygen enriched air). The air inlet stream may be heated in the air recuperator **312** to a temperature above 500° C., such as 550° C. to 600° C. The heated air inlet stream is provided from the air recuperator **312** to the air inlet of the stack **100** via the stack air inlet conduit **212**.

[0054] The oxygen exhaust is output from the air recuperator **312** to the water preheater **302** via the oxygen exhaust conduit **228** at temperature above 200° C., such as 250° C. to 350° C. The oxygen exhaust is output from the water preheater **302** via the exhaust conduit **205** at temperature of 100° C. to 150° C.

[0055] According to various embodiments, the system **300** may include an optional air preheater heat exchanger **354** disposed outside or inside of the hotbox **301**. In particular, the air preheater **354** may be configured to preheat the air inlet stream provided to the hotbox **301** by the air blower **318** via the air inlet conduit **218** using heat in the hydrogen exhaust (i.e., the hydrogen and steam product stream) from the stack **100**. The air may be preheated in the air preheater **354** to a temperature above 100° C., such as 150° C. to 250° C. The hydrogen exhaust may be provided from the steam recuperator **308** to the air preheater **354** via a hydrogen conduit **238**. In an optional embodiment, the ejector **400** may allow the steam recycle temperature in the hydrogen recycle manifold **210A** to be higher than in a system where the steam recycle is provided by a blower. Thus, the ejector **400** permits use of a higher temperature steam if it is available, compared to a using a recycle blower.]

[0056] The air heater **355** may be located radially outward of the air recuperator **312** in the hotbox **301**. The air heater **355** may further heat the air inlet stream flowing through the air recuperator **312** and/or the stack air inlet conduit **312**. The stack heater **360** may be located radially inward of the stacks **100** or columns and radially outward of the steam recuperator **308** in the hotbox **301**. The stack heater **360** may heat the stacks **100** or columns. The steam heater **370** may be located radially inward of the steam recuperator **308** in the center of the hotbox **301**. The steam heater **370** may heat the steam inlet stream flowing through the steam recuperator **308** and/or the fuel inlet conduit

208.

[0057] The stack heater **360** may have two or more heating regions. For example, the stack heater **360** may include a lower heating region and an upper heating region. Heating elements included in each of the regions may be independently connected to a voltage source. The heating elements of each region may be independently controlled by the system controller **325**, in order to independently control an amount of heat generated by each heating region. As such, the stack heater **360** may radiate different amounts of heat to different portions of the stacks **100** or columns **101** and/or different portions of the central column **320**. The system may include temperature sensors to monitor the various zones in order to provide feedback to the system controller **325** so that it can adjust the amount of heating provided to the different regions accordingly.

[0058] According to various embodiments, the system **300** may include a system controller **325**, such as a central processing unit, that is configured to control the operation of the system **300**. For example, the controller **325** may be wired or wirelessly connected to various elements of the system **300** to control the same and communicate with other components connected to system **300**. The optional upstream pressure regulator **62** and/or optional downstream pressure regulator **401** may be provided on the steam and hydrogen conduit **206** upstream and/or downstream of the fluid ejector **400**, respectively. The upstream pressure regulator **62** may be omitted if the system site fuel pressure is controlled to be in a desired range.

[0059] FIG. **4** is a schematic diagram of a fluid ejector **400** that may be utilized in systems **200** or **300** of FIGS. **2** and **3**, according to various embodiments of the present disclosure. Referring to FIG. **4**, the fluid ejector **400** may include a suction chamber **402**, a mixing chamber **404**, and a diffusion chamber **406** that are fluidly connected. The suction chamber **402** may include an inlet nozzle (e.g., the primary inlet) **410** and the secondary inlet **412**, and the diffusion chamber may include an outlet **414**.

[0060] The inlet nozzle **410** may be configured to inject a primary fluid stream into the suction chamber **402**. The primary fluid stream may comprise a fuel stream of fresh fuel (e.g., hydrogen) provided from the fuel inlet **102** of FIG. **2**, or a steam inlet stream provided from the steam generator **304** of FIG. **3** (or mixer **306** as the case may be). The inlet nozzle **410** may be configured to increase the velocity of and/or reduce the pressure of the primary fluid stream, such that the primary fluid stream is injected to the suction chamber **402** at a higher velocity and a lower pressure than the primary fluid stream is provided to the inlet nozzle **410**. The relatively high velocity of the primary fluid stream may reduce the pressure in the suction chamber **402**.

[0061] The secondary inlet **412** may be configured to provide a secondary fluid stream to the suction chamber **402**. For example, the secondary fluid stream may comprise the anode exhaust stream received via the manifold **110** from the hotbox **201** of FIG. **2** or a hydrogen exhaust stream received via the manifold **210A** from the hotbox **301** of FIG. **3**. The primary fluid stream may be provided to the inlet nozzle **410** at a higher pressure than the secondary fluid stream is provided to the secondary inlet **412**. Within the suction chamber **402**, the velocity of the secondary fluid stream may be increased due to interaction with the high-velocity, turbulent, primary fluid stream.

[0062] In some embodiments, the internal cross-sectional area and/or diameter of the suction chamber **402** may decrease adjacent to the mixing chamber **404**, which may further accelerate both the primary and secondary fluid streams. For example, the internal cross-sectional area of the suction chamber **402** may decrease as the distance to the mixing chamber **404** decreases. In one embodiment, the diffusion chamber **406** comprises a diverging nozzle, and the mixing chamber **404** is narrower (e.g., has a smaller diameter) than the diffusion chamber **406** and the suction chamber **402**.

[0063] The primary and secondary fluid streams may flow into the mixing chamber **404** where the fluid streams are mixed to form a mixed stream. The mixed stream may then flow into the diffusion chamber **406**. The diffusion chamber **406** may be configured to reduce the velocity and/or increase the pressure of the mixed stream. For example, the internal cross-sectional area and/or diameter of

the diffusion chamber **406** may increase as the distance from the mixing chamber **404** increases. Accordingly, the mixed stream may be provided to the outlet **414** of the diffusion chamber **406** at a pressure and/or velocity that is suitable for operation of an electrochemical cell stack.

[0064] Accordingly, the fluid ejector **400** may operate based on Bernoulli's principle, wherein as the velocity of a fluid through a conduit increases, its pressure decreases, and vice versa. For example, the reduced pressure generated in the suction chamber **402** by the inlet nozzle **410** may generate suction that pulls the secondary fluid stream into the fluid ejector **400**. The fluid ejector **400** may also reduce the velocity and/or increase the pressure of the fuel mixture output to the stack **100**.

[0065] In other electrochemical cell systems, a recycle blower is needed to maintain a required stack inlet pressure. However, recycle blowers consume power and require relatively complicated moving parts. In addition, blowers may struggle with the high temperatures present in high-temperature fuel cell operating conditions. In contrast, the fluid ejector **400** may allow for the omission of a recycle blower, and thereby improve system power efficiency and reliability, while also reducing manufacturing and maintenance costs.

[0066] FIG. 5A is a schematic diagram showing a process flow in a fuel cell system **500**, such as a SOFC system, according to various embodiments of the present disclosure. The system **500** may be similar to the system **200** of FIG. 2. Accordingly, only the differences therebetween will be discussed in detail.

[0067] Referring to FIGS. 2, 4, and 5A, the system **500** may be configured to operate using hydrogen as a fuel and may include multiple power modules **202** and corresponding components housed therein. The power modules **202** may be fluidly connected to a secondary inlet of a fluid ejector **400** by a recycling manifold **510** or conduit (e.g., an anode exhaust recycling manifold **110** shown in FIG. 2). The power modules **202** may be fluidly connected to an outlet **414** of the fluid ejector **400** by an inlet manifold **520** or conduit (e.g., fuel manifold). A condenser **180** may be disposed on the recycling manifold **510**, such that a hydrogen outlet of the condenser **180** is fluidly connected to a secondary inlet **412** of the fluid ejector **400**. The water outlet of the condenser **180** may be connected to a water drain. An inlet nozzle **410** of the fluid ejector **400** may be fluidly connected to a fuel inlet conduit **42** or manifold. The fuel inlet conduit **42** may be fluidly connected to the fuel inlet **102**, such as a hydrogen storage vessel or a hydrogen line. An isolation valve **60** and an optional pressure regulator **62** may be located on the fuel inlet conduit **42**. The upstream pressure regulator **62** upstream of the fluid ejector **400** may be omitted if the downstream pressure regulator **401** is located downstream of the fluid ejector **400** and/or if the system site fuel pressure is controlled to be in a desired range.

[0068] Accordingly, the condenser **180** may operate to remove water from anode exhaust generated by fuel cell stacks **100** disposed in multiple power modules **202**. The fluid ejector **400** may operate to pull the anode exhaust through the condenser **180** and/or the recycling manifold **510**.

[0069] FIG. 5B is a schematic diagram showing a process flow in an alternative SOFC system **500A**, according to various embodiments of the present disclosure. The system **500A** may be similar to the system **500**. Accordingly, only the differences therebetween will be discussed in detail.

[0070] Referring to FIG. 5B, the system **500A** may include multiple fluid ejectors **400** to maintain a desired pressure in a fuel inlet stream provided to the power modules **202**. For example, the system **500A** may include a first fluid ejector **400A**, a second fluid ejector **400B**, a first nozzle valve **64A**, a second nozzle valve **64B**, a first inlet valve **66A**, and a second inlet valve **66B**. During steady-state operations, the valves **64A**, **64B**, **66A**, **66B**, may be opened to provide sufficient fuel and anode exhaust flow through the manifolds **510**, **520**. In particular, both fluid ejectors **400A**, **400B** may be operated to provide a mixed stream comprising fresh fuel (e.g., hydrogen) and anode exhaust to the power modules **202** via the inlet manifold **520**.

[0071] However, one or more of the power modules **202** may be taken offline during servicing. As

such, the fuel and anode exhaust flow rates of the system **500A** may be correspondingly reduced. Therefore, one of the fluid ejectors **400A**, **400B** may be isolated from the fuel and anode exhaust flows, by closing the corresponding valves **64A** and **66A** or **64B** and **66B**.

[0072] FIG. **5C** is a schematic diagram showing a process flow in an alternative SOFC system **500B**, according to various embodiments of the present disclosure. The system **500B** may be similar to the system **500**. Accordingly, only the differences therebetween will be discussed in detail.

[0073] Referring to FIG. **5C**, the system **500B** may include a recycle blower **190**, an inlet valve **66**, and a blower valve **68**. An inlet of the recycle blower **190** may be fluidly connected to an outlet of the condenser **180** by the recycling manifold **510**. The recycling manifold **510** may include a bypass conduit **510A** that fluidly connects an outlet of the recycle blower **190** directly to the inlet manifold **520**. In other words, the bypass conduit **510A** may bypass the fluid ejector **400**, such that at least a portion of the anode exhaust is provided directly to the inlet manifold **520**, downstream of the fluid ejector **400** with respect to a fluid flow direction through the inlet manifold **520**. The blower valve **68** may be configured to selectively close the bypass conduit **510A**.

[0074] During steady-state operation, the inlet valve **66** may be open and the blower valve **68** may be closed, such that all of the anode exhaust output from the power modules **202** is provided to the fluid ejector **400**. During non-steady state and/or transition operating modes, the blower valve **68** may be opened and the recycle blower **190** may be operated to bypass the fluid ejector **400** and/or increase anode exhaust flow to the inlet manifold **520**. The inlet valve **66** may be closed to prevent fuel backflow from the fluid ejector **400** into the recycle blower **190**. In some embodiments, the system **500B** may include additional blowers **190** and corresponding valves **66**, **68**. The coordination of the opening and closing of the various valves during different operating modes can be accomplished by the system controller **125**.

[0075] FIG. **5D** is a schematic diagram showing a process flow in an alternative SOFC system **500C**, according to various embodiments of the present disclosure. The system **500C** may include a combination of the components of the systems **500A** and **500B**. Accordingly, only the differences therebetween will be discussed in detail.

[0076] Referring to FIG. **5D**, the system **500C** may include at least one condenser **180**, at least two fluid ejectors **400A**, **400B**, at least one recycle blower **190**, and corresponding valves **64A**, **64B**, **66A**, **66B**, **68** to control fuel and anode exhaust flow therethrough. Accordingly, fluid flows through the fluid ejectors **400A**, **400B**, and the recycle blower **190** may be independently controlled.

[0077] FIG. **6A** is a schematic diagram showing a process flow in a SOEC system **600**, according to various embodiments of the present disclosure. The system **600** may be similar to the system **300** of FIG. **3**. Accordingly, only the differences therebetween will be discussed in detail.

[0078] Referring to FIG. **6A**, the system **600** may include multiple fluid ejectors **400** to maintain a desired pressure in a steam stream provided to the hydrogen generation module **302**. For example, the system **600** may include a first fluid ejector **400A**, a second fluid ejector **400B**, a first nozzle valve **64A**, a second nozzle valve **64B**, a first inlet valve **66A**, and a second inlet valve **66B**. The hydrogen generation module **302** may be fluidly connected to the fluid ejectors **400A**, **400B** by a recycling manifold **610** or conduit and an inlet manifold **620** or conduit (e.g., steam manifold). In particular, the recycling manifold **610** may be configured to provide hydrogen and/or water output from the hydrogen generation module **302** to the fluid ejectors **400A**, **400B** and/or to a hydrogen processor **320**. In particular, the recycling manifold **610** may include an orifice **616** and a hydrogen flow divider **322**, such as a splitter and/or valve, configured to divert at least a portion of the generated hydrogen stream to the fluid ejectors **400A**, **400B** during steady-state operation while the remainder of the generated hydrogen stream is provided to the hydrogen processor **320**. The optional orifice **616** is located downstream from the hydrogen flow divider **322** and is sized based on inlet pressure needed to meet the desired flow rate. The orifice **616** is used to limit the flow of

the recycled hydrogen to the fluid ejectors **400A**, **400B**. Alternatively, the orifice **616** may be omitted.

[0079] During steady-state operation, the valves **64A**, **64B**, **66A**, **66B**, may be opened to provide sufficient fuel and anode exhaust flow through the manifolds **610**, **620**. In particular, both fluid ejectors **400A**, **400B** may be operated to provide a mixed steam comprising steam and hydrogen to the hydrogen generation module **302** via the inlet manifold **620**.

[0080] In some embodiments, hydrogen and steam flow to at least one of the fluid ejectors **400A**, **400B** may be reduced or stopped by closing or partially closing the corresponding valves **64A**, **64B**, **66A**, **66B**. For example, hydrogen and steam flow may be reduced or stopped during a transition operation, servicing, and/or during low-load steady state operation.

[0081] FIG. **6B** is a schematic diagram showing a process flow in an alternative SOEC system **600A**, according to various embodiments of the present disclosure. The system **600A** may be similar to the system **300** of FIG. **3**. Accordingly, only the differences therebetween will be discussed in detail.

[0082] Referring to FIG. **6B**, the system **600A** may include a recycle blower **190**, an inlet valve **66**, and a blower valve **68**. The recycle blower **190** may be fluidly connected to the recycling manifold **610** and the inlet manifold **620**. In particular, the recycling manifold **610** may include a bypass conduit **610A** that fluidly connects an outlet of the recycle blower **190** directly to the inlet manifold **620**, downstream of the fluid ejector **400** with respect to a fluid flow direction through the inlet manifold **620**. The blower valve **68** may be configured to selectively close the bypass conduit **610A**.

[0083] During non-steady state and/or transition operations, the blower valve **68** may be opened and the recycle blower **190** may be operated to bypass the fluid ejector **400** and increase hydrogen flow to the inlet manifold **620**. The inlet valve **66** may be closed or partially closed to prevent backflow from the fluid ejector **400** into the recycle blower **190**. In one embodiment, the recycle blower **190** may have a longer life span compared to a comparative embodiment where the ejector **400** is omitted, because the recycle blower **190** operates intermittently (e.g., during non-steady or transition operations) in the embodiment in which the ejector **400** is present, compared to being operated continuously during system operation in the comparative embodiment in which the ejector **400** is omitted.

[0084] FIG. **6C** is a schematic view showing a process flow in an alternative SOFC system **600B**, according to various embodiments of the present disclosure. The system **600B** may include a combination of the components of the systems **600** and **600A**. Accordingly, only the differences therebetween will be discussed in detail.

[0085] Referring to FIG. **6C**, the system **600B** may include at least two fluid ejectors **400A**, **400B**, a recycle blower **190**, and the corresponding valves **64A**, **64B**, **66A**, **66B**, **68** to control steam and hydrogen flow therethrough. Accordingly, fluid flow through the fluid ejectors **400A**, **400B**, and the recycle blower **190** may be independently controlled by adjusting the valves **64A**, **64B**, **66A**, **66B**, **68**, and/or controlling the operation of the recycle blower **190**.

[0086] The optional pressure regulator **62** in the systems **600**, **600A** and/or **600B** of FIGS. **6A**, **6B** and/or **6C** may be replaced with a flow control valve (e.g., a valve similar to the valve **251** shown in FIG. **3** which controls the steam flow rate) if the system site control pressure of the steam is regulated. Alternatively, a flow control valve may be used in fluid series with the pressure regulator **62**.

[0087] FIG. **7A** is a schematic diagram showing a process flow in an SOEC system **700**, according to various embodiments of the present disclosure. The system **700** may be similar to the system **300** of FIG. **3**. Accordingly, only the differences therebetween will be discussed in detail.

[0088] Referring to FIGS. **3**, **4**, and **7A**, the system **700** may include multiple hydrogen generation modules **302** and at least one fluid ejector **400**. The hydrogen generation modules **302** may be fluidly connected to a secondary inlet of the fluid ejector **400** by a recycling manifold **710** (e.g., a

hydrogen manifold) or conduit and may be fluidly connected to an outlet **414** of the fluid ejector **400** by an inlet manifold **720** or conduit (e.g., steam manifold).

[0089] The recycling manifold **710** may be configured to provide hydrogen and/or water output from the hydrogen generation modules **302** to the secondary inlet **412** the fluid ejector **400** and/or to a hydrogen processor **320**. In particular, the recycling manifold **710** may include an orifice **716** and a hydrogen flow divider **322**, as described above. The system **700** may also include an inlet manifold **720** configured to provide steam and hydrogen from an outlet **414** of the fluid ejector **400** to the hydrogen generation modules **302**.

[0090] FIG. 7B is a schematic diagram showing a process flow in an alternative SOEC system **700A**, according to various embodiments of the present disclosure. The system **700A** may be similar to the system **700** of FIG. 7A. Accordingly, only the differences therebetween will be discussed in detail.

[0091] Referring to FIG. 7B, the system **700A** may include a recycle blower **190**, an inlet valve **66**, and a blower valve **68**. The recycle blower **190** may be fluidly connected to the recycling manifold **710** and the inlet manifold **720**. In particular, the recycling manifold **710** may include a bypass conduit **710A** that fluidly connects an outlet of the recycle blower **190** directly to the inlet manifold **720**, downstream of the fluid ejector **400** with respect to a fluid flow direction through the inlet manifold **720**. The blower valve **68** may be configured to selectively close the bypass conduit **710A**.

[0092] During non-steady state and/or transition operations, the blower valve **68** may be opened and the recycle blower **190** may be operated to bypass the fluid ejector **400** and increase hydrogen flow to the inlet manifold **720**. The inlet valve **66** may be closed or partially to prevent backflow from the fluid ejector **400** into the recycle blower **190**.

[0093] FIG. 7C is a schematic diagram showing a process flow in an alternative SOEC system **700B**, according to various embodiments of the present disclosure. The system **700B** may be similar to the system **700** of FIG. 7A. Accordingly, only the differences therebetween will be discussed in detail.

[0094] Referring to FIG. 7C, the system **700B** may include multiple fluid ejectors **400** to maintain a desired pressure in a steam stream provided to the multiple hydrogen generation modules **302**. For example, the system **600** may include a first ejector **400A**, a second ejector **400B**, a first nozzle valve **64A**, a second nozzle valve **64B**, a first inlet valve **66A**, and a second inlet valve **66B**. The hotboxes **302** may be fluidly connected to the fluid ejectors **400A**, **400B** by a recycling manifold **710** or conduit and an inlet manifold **720** or conduit. In particular, the recycling manifold **710** may be configured to provide hydrogen and/or water output from the hotbox **101** to the fluid ejectors **400A**, **400B** and/or to downstream process such as a hydrogen processor. In particular, the recycling manifold **710** may include diverters **716**, such as a splitter, pump, blower, orifice, and/or valve, configured to selectively divert at least a portion of the generated hydrogen stream to the fluid ejectors **400A**, **400B** during steady-state operation.

[0095] During steady-state operation, the valves **64A**, **64B**, **66A**, **66B**, may be opened to provide sufficient fuel and anode exhaust flow through the manifolds **710**, **720**. In particular, both fluid ejectors **400A**, **400B** may be operated to provide a mixed steam comprising steam and hydrogen to the hotboxes **302** via the inlet manifold **720**.

[0096] In some embodiments, hydrogen (H.sub.2) and steam flow to at least one of the fluid ejectors **400A**, **400B** may be reduced or stopped by closing or partially closing the corresponding valves **64A**, **64B**, **66A**, **66B**. For example, hydrogen and steam flow may be reduced during transition operation and/or during low-load steady state operation.

[0097] FIG. 7D is a schematic diagram showing a process flow in an alternative SOEC system **700C**, according to various embodiments of the present disclosure. The system **700C** may include a combination of the components of the systems **700A** and **700B**. Accordingly, only the differences therebetween will be discussed in detail.

[0098] Referring to FIG. 7D, the system 700C may include at least two fluid ejectors 400A, 400B, a recycle blower 190, and the corresponding valves 64A, 64B, 66A, 66B, 68 to control steam and hydrogen flow therethrough. Accordingly, fluid flows through the fluid ejectors 400A, 400B, and the recycle blower 190 may be independently controlled by adjusting the valves 64A, 64B, 66A, 66B, 68, and/or controlling the operation of the recycle blower 190.

[0099] The optional pressure regulator 62 in the systems 700, 700A, 700B and/or 700C of FIGS. 7A, 7B, 7C and/or 7D may be replaced with a flow control valve (e.g., a valve similar to the valve 251 shown in FIG. 3 which controls the steam flow rate) if the system site control pressure of the steam is regulated. Alternatively, a flow control valve may be used in fluid series with the pressure regulator 62.

[0100] The fuel cell fuel cell systems of various embodiments of the present disclosure are designed to reduce greenhouse gas emissions and have a positive impact on the climate.

[0101] The preceding description of the disclosed aspects is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects without departing from the scope of the invention. Thus, the present invention is not intended to be limited to the aspects shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. An electrochemical cell system, comprising: a stack of electrochemical cells; a first fluid ejector configured to mix a first fluid stream and a second fluid stream to form a mixed fluid stream that is provided to the stack, the first fluid ejector comprising: a suction chamber comprising a secondary inlet configured to receive the second fluid stream; an inlet nozzle configured to inject the first fluid stream into the suction chamber; a mixing chamber fluidly connected to the suction chamber and configured to mix the first fluid stream and the second fluid stream to form the mixed fluid stream; and a diffusion chamber fluidly connected to the mixing chamber and comprising an outlet that is fluidly connected to the stack.
2. The electrochemical system of claim 1, wherein: the first fluid stream is configured to be provided to the inlet nozzle at a first velocity and a first pressure; and the inlet nozzle is configured to inject the first fluid stream into the suction chamber at a second velocity that is higher than the first velocity and at a second pressure that is lower than the first pressure.
3. The electrochemical cell system of claim 2, wherein the suction chamber is configured to increase a velocity of the second fluid stream.
4. The electrochemical cell system of claim 2, wherein the diffusion chamber is configured to output the mixed fluid stream from the outlet at a third velocity that is less than the second velocity and at a third pressure that is higher than the second pressure.
5. The electrochemical cell system of claim 1, wherein the diffusion chamber comprises a diverging nozzle and the mixing chamber is narrower than the diffusion chamber and the suction chamber.
6. The electrochemical cell system of claim 1, further comprising: a supply manifold or conduit configured to provide the first fluid stream to the inlet nozzle from a fluid source; a recycling manifold or conduit fluidly connected to an outlet of the stack, and configured to provide the second fluid stream from the stack to the secondary inlet; and an inlet manifold or conduit fluidly connecting the outlet to the stack.
7. The electrochemical cell system of claim 6, further comprising a second fluid ejector fluidly connected to the supply manifold or conduit, the recycling manifold or conduit, the inlet manifold or conduit, and the stack.
8. The electrochemical cell system of claim 6, further comprising a recycle blower fluidly connected to the recycling manifold or conduit and configured to provide the second fluid stream

from the recycling manifold or conduit to the stack via the inlet manifold or conduit while bypassing the first fluid ejector.

9. The electrochemical cell system of claim 6, further comprising additional stacks of electrochemical cells located in a plurality of module housings.

10. The electrochemical cell system of claim 9, wherein: the recycling manifold or conduit is configured to provide the second fluid stream from the plurality of module housings to the secondary inlet; and the inlet manifold or conduit fluidly connects the outlet to the stacks located in the plurality of module housings.

11. The electrochemical cell system of claim 1, wherein: the inlet nozzle is fluidly connected to a hydrogen fuel source; the first fluid stream comprises a hydrogen fuel; the second fluid stream comprises hydrogen and water; and the electrochemical cells comprise fuel cells.

12. The electrochemical cell system of claim 11, further comprising a condenser configured to remove the water from the second fluid stream before the second fluid is provided to the first fluid ejector.

13. The electrochemical cell system of claim 1, wherein: the inlet nozzle is fluidly connected to a steam source; the first fluid stream comprises steam; the second fluid stream comprises hydrogen and water; and the electrochemical cells comprise electrolyzer cells.

14. A method of operating an electrochemical cell system, comprising: providing a first fluid stream to an inlet nozzle of a fluid ejector; providing a second fluid stream from a stack of electrochemical cells to a second inlet located in a suction chamber of the fluid ejector; mixing the first fluid stream and the second fluid stream in a mixing chamber of the fluid ejector to form a mixed fluid stream; and providing the mixed fluid stream from a diffusion chamber of the fluid ejector to the stack.

15. The method of claim 14, wherein: the first fluid stream is provided to the inlet nozzle at a first velocity and a first pressure; the inlet nozzle injects the first fluid stream into the suction chamber at a second velocity that is higher than the first velocity and at a second pressure that is lower than the first pressure; the suction chamber increases a velocity of the second fluid stream; and the diffusion chamber outputs the mixed fluid stream from the outlet at a third velocity that is less than the second velocity and at a third pressure that is higher than the second pressure.

16. The method of claim 15, wherein the diffusion chamber comprises a diverging nozzle and the mixing chamber is narrower than the diffusion chamber and the suction chamber.

17. The method of claim 14, wherein: the first fluid stream comprises hydrogen fuel; the second fluid stream comprises hydrogen and water; and the electrochemical cells comprise fuel cells.

18. The method of claim 17, further comprising removing the water from the second fluid stream before the second fluid is provided to the fluid ejector.

19. The method of claim 14, wherein: the first fluid stream comprises steam; the second fluid stream comprises hydrogen and water, and the electrochemical cells comprise electrolyzer cells.

20. The method of claim 14, further comprising using a recycle blower in a non-steady state mode to provide the second fluid stream to the stack while bypassing the fluid ejector.
